

A Singular Dynamic AMM Pricing American-Style Perpetual Options Across the Strike Continuum

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Abstract. Perpetual options let traders cap liquidation risk without closing their positions. We price the whole out-of-the-money strike continuum, calls and puts, in closed form from a *single* weighted constant-product (“Balancer”) liquidity pool—not the generalized-hyperbolic curve of prior work—with one weight the pool updates per trade. The curve family is not the contribution; the new ideas are that weight update, a layer pricing the continuum from one pool state, a lift of the prior *barrier* instrument into an *American-style* option with a closed-form early-exercise boundary (continuous in value and slope, no new parameters) tied to the classical perpetual-American (Merton) problem on the Gaussian case, and a static, volatility-calibrated curvature knob fixing the option-value steepness uniformly across strikes. The boundary, the no-arbitrage symmetry, and the curve/knob spine are machine-checked by the Aristotle prover (*trusted-from-prover*: trusted as compiled, not re-run against a local kernel). The conditions are stated up front: “American” is proved over *deterministic* exercise rules, the stochastic optimal-stopping identification named but not formalised; the any-asset extension rests on distributional hypotheses carried, not discharged; replication is exact only per wing for hedge ratio at most one (beyond, a labelled approximation); and *solvency* stays conditional on a funding source shown necessary but not discharged.

Keywords: Automated market maker · Decentralized finance · Perpetual options · Optimal stopping · Smooth pasting · Formal verification

1 Introduction and contributions

A perpetual future lets a trader hold leveraged directional exposure with no expiry, at the cost of a discontinuous, ruinous *liquidation* when collateral is exhausted. A perpetual option on the same underlying—a call to hedge a short, a put to hedge a long—caps realised loss beyond a chosen strike and, because it does not expire, holds that protection for as long as the underlying perp is held. The pricing of such a continuum of strikes is the problem this paper addresses.

The result, stated first. Our central positive result is that a *single* liquidity pool prices the entire out-of-the-money strike continuum—calls and puts—of

American-style perpetual options in *closed form*. The position’s holder exercises not at the strike but at an exact *early-exercise free boundary*: the price level $S^* = K\gamma/(\gamma + 1)$ at which it becomes optimal to take the payoff, written in terms of the dollar strike K and the curve’s *convexity exponent* $\gamma > 1$ (the larger γ is, the steeper the option-value curve). The continuation value meets the exercised value there with matching value *and* slope—a *smooth* join we call the *seam* (C^1 , i.e. once continuously differentiable)—and the formula needs no new parameters. For the shipped instrument—a vanilla opposite-wing *band* (one bought leg, one sold leg, at most one in-the-money at a time)—the construction is an *exact* per-wing replication of the target payoff, and the boundary sits on a clean tie to the classical perpetual-American (Merton) problem: γ turns out to be the natural exponent of the asset’s price process (its “characteristic root”), and S^* is exactly the Merton early-exercise level when the asset is taken to be log-normal (the *Gaussian slice*). That is the strength; we are equally explicit about its two conditions. First, the Merton/Gaussian anchor is exact only on that Gaussian slice: extending the boundary to an *arbitrary* asset rests on a distributional layer (a generalized hyperbolic / Bessel-K family) that we carry as *named hypotheses*, not as discharged facts. Second, “American” here means optimal over *deterministic* exercise boundaries (a genuine variational result); the identification with the *stochastic* optimal-stopping (Snell) boundary is *named but not formalised*—a placeholder obligation, not a proven premise. The two further honest bounds—exact replication only for hedge-ratio magnitude $|I| \leq 1$ (the position’s sensitivity to spot, capped at one per wing), and *conditional* solvency—are stated below.

Scope and standing assumptions. So that no condition reads as a hidden defect, we state the principal one plainly and up front: *solvency in this work is conditional*. It holds only under named hypotheses—B1 (a funding port covers the concave deficit), B3 (arbitrage is one-way), and B4 (the settlement ledger closes)—which are carried as *structure fields*, not discharged facts. The construction proves “*if* the ledger closes, arbitrage is one-way, and the port covers the floor, *then* the pool is passive and solvent”; it does *not* prove that the port pays in the real world. Geometry is shown *necessary* for solvency (*reserves_have_no_floor*) but never sufficient. These named hypotheses are therefore assumptions, not claims, and the reader should not mistake the conditional solvency result for an unconditional one (Sections 8, 10).

An earlier barrier-option treatment [1] priced the OTM continuum with a single weighted constant-product pool and treated each position as a *barrier* instrument: its value rises toward one perpetual-future unit as spot approaches the strike, then is capped at that ceiling once in-the-money and held there. This draft keeps that entire pricing spine and *lifts* the barrier ceiling into an American-style *early-exercise free boundary*. The ceiling becomes the point at which optimal exercise occurs; continuation runs *past* the strike to a formula-determined boundary and then pays intrinsic.

Contributions, and where the novelty is not. We are explicit, following a careful prior-art deflation, about what is and is not new. The pool holds reserves of two assets, x of the underlying and y of cash. The *curve family* is not the contribution: if one slides the origin to *shifted virtual reserves* $u = x - \alpha$ and $v = y - \beta$ (subtracting offsets α, β defined in Section 3), each individual trade’s arc is an ordinary constant product $u \cdot v = \text{const}$, so the geometry per se is a translated constant-function market maker (CFMM, an automated market maker whose reserves obey a fixed curve) in the sense of Evans [8] and the replication theory of Angeris–Evans–Chitra [9]. The contributions are:

1. A *per-trade endogenous weight update* on the Balancer invariant $x^w y^{1-w} = k$ —the rule that the weighted geometric mean of the two reserves stays on a fixed curve, where the *weight* $w \in (0, 1)$ sets how the curve leans between the two assets and k is the curve’s level. It prices the whole out-of-the-money (OTM) call/put strike continuum in closed form, serving a tradable per-strike claim. The pool is a *state-transition system* (its behaviour is a sequence of state updates, one per trade, not a single fixed equation): the weight w is genuine stored state, the conservation law is the per-trade rule applied at the trade’s own point, and the invariant level k is a dependent readout, not a conserved quantity (Section 3).
2. A *semantic layer* over that pool: strike-as-ray with a unique trade point per ray, the composite-ray identity for settlement and its cross-wing obstruction, per-strike funding referenced to the anchor curve, and a carved-perp settlement denomination (Sections 3, 6).
3. The *American generalisation*: a closed-form smooth-pasting free boundary $S^* = K \gamma / (\gamma + 1)$ with C^1 value/slope match and no new parameters, replacing the barrier ceiling (Section 5). The boundary and its C^1 join are machine-checked by the Aristotle prover (*trusted-from-prover*).
4. A vol-calibrated *constant slope-multiplier* lens m acting as a kurtosis knob: it fixes the option-value steepness to $m\gamma$ *uniformly across strikes*, with $m = 1$ recovering the plain Balancer curve (Section 4). The pool flow itself is unchanged by the lens. This is the *single static, volatility-calibrated curvature parameter set at pool creation* that a vol-free mark otherwise lacks (Section 4 positions it as such); it gives the funding mechanism a vol-aware scale.
5. A structural no-internal-arbitrage result: a costless collar yields zero capital-efficiency surplus *if and only if* the pool is symmetric ($w = \frac{1}{2}$), a genuine biconditional with an explicit skew counterexample (Section 7), machine-checked by Aristotle (*trusted-from-prover*).
6. An honest *formal-verification ledger* (Section 8) that states exactly what is machine-checked, under what provenance label, and what remains the open frontier—namely, that *solvency is conditional* on a funding port shown necessary but not discharged.

A note on honesty of claims. Throughout, every formal claim is tagged with its provenance. “*trusted-from-prover*” means the statement was compiled by the Aristotle theorem prover server-side in the matching toolchain (Lean 4.28.0 /

Mathlib v4.28.0) and passed a zero-cost artifact audit (forbidden-token scan, prover-reported `#print axioms` $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, unscoped-module diff, hand re-derivation), but was *not* re-run against a local canonical kernel here. We never write “verified” for these results: that label is reserved for a local canonical-env build, which is environment-blocked in the present work. Where a claim is a conjecture or a research direction, it is labelled as such and never presented as proven.

2 American-style perpetual options

2.1 Definition

Perpetual options are non-expiring contracts-for-difference representing the pay-off from favourable price action—upward for calls, downward for puts—of an underlying asset relative to a denominating asset (USD here) beyond a strike K . In the prior draft the instrument was a *barrier*: its value topped out at one perpetual-future unit when in-the-money (ITM) and stayed at that ceiling. Here the instrument is *American-style*: while the holder remains out-of-the-money the position carries continuation value; once spot moves far enough in the holder’s favour, optimal exercise pays the intrinsic value measured from the strike. The transition is not at the strike itself but at a free boundary determined below (Section 5).

2.2 Utility

Liquidation in a perpetual future is the involuntary termination of a leveraged position when collateral is exhausted. Composing a perp with an OTM perpetual option on the same underlying caps realised loss past the strike, bounding the trader’s drawdown without forcing position closure; because the option does not expire, the hedge persists for the life of the perp. The American generalisation makes the protected leg behave like a genuine early-exercisable claim rather than a hard barrier, which is what a perpetual liquidation hedge economically is: there is no expiry to wait for, so the holder’s only decision is *when* the protection is best realised.

2.3 Pricing: the mark, with no separate premium

For an OTM strike the value of a position is its *mark*: the lesser of the strike-ray slope and its reciprocal,

$$\text{mark} = \min(\text{slope}, 1/\text{slope}) \in (0, 1], \quad (1)$$

a dimensionless quantity read as the *fraction of a full perpetual future* the position is worth: it approaches 1 near the money and decays toward 0 deep OTM. The value of a position of size q is $q \cdot \text{mark}$.

This pricing carries *no separate premium object*. The mark is the complete description of what a position is worth; there is no extrinsic/intrinsic decomposition layered on top, and “the value of a position” means $q \cdot \text{mark}$ and nothing else. The American generalisation does not introduce a premium either: it only changes how the mark is computed once spot moves past the strike (continuation through the strike, then intrinsic), as Section 5 makes precise.

3 The AMM construction

We make two generalisations to the constant-function AMM $x^w y^{1-w} = k$ that give it the expressivity to price perpetual options across the continuum. Both are inherited unchanged from the prior draft.

3.1 Trades skew the curve

“Trades skew the AMM curve instead of moving the reserves point along it.” A two-asset constant-function market maker (CFMM) is a curve in the (x, y) plane along which the reserves (x, y) are a single point; the price is the slope there, and an ordinary trade slides the reserves point along the curve.

Two curves are in play. The **Balancer curve** $x^w y^{1-w} = k$ has its *shape* set by the weight w ; this is the curve that skews. For the *spot-trade operator only* (a swap executed at the reserves point), the reachable states trace the **trajectory hyperbola**

$$(x - \alpha)(y - \beta) = \alpha\beta, \quad \alpha = xw, \quad \beta = y(1 - w), \quad (2)$$

which is tangent to the Balancer curve at the reserves point—they share a slope there—so spot pricing read off the Balancer curve is faithful, and “reshaping the curve” and “moving along the trajectory” describe the same spot event from two angles. We stress the scope: (2) is the reachable set of the *spot* operator. The off-ATM perpetual-option trade (next subsections) is a *distinct* operator with its own local hyperbola through the trade point, and does not move the global reserves point along (2).

The prior-art deflation, stated plainly. Writing $u = x - \alpha_T$, $v = y - \beta_T$ for the conserved pair at the trade point T , each trade’s arc satisfies $uv = \alpha_T \beta_T$: an ordinary constant product on *shifted virtual reserves*. The curve geometry is therefore not new—it is a translated CFMM in the sense of Evans [8], and the “reshaping” picture is a re-description of a per-trade weight change, not a novel invariant. What is new is that the weight is updated per trade to serve a tradable per-strike claim, and the semantic layer built on top; see Section 9.

3.2 Trades happen anywhere on the curve

In ordinary CFMMs trades are spot swaps, always at the reserves point. We treat spot swaps as the special case of perpetual-option swaps, which may occur

at *any* OTM point on the curve. Each ray from the origin—a line of fixed reserve ratio y/x , whose slope we denote θ —is identified with a unique strike; the ray meets the curve at one *trade point* (its single intersection with the curve), and a transaction at that point is handled as if the trade point were the reserves point. The 45° ray ($y = x$, i.e. $\theta = 1$) is the at-the-money strike; rays steeper and shallower partition the OTM continuum into the two *wings*—the call side and the put side of the strike continuum—both priced from the same (x, y, w) state.

3.3 The trade as a state transition (conservation applied at the trade's own point)

The pool is a state transition system, *not* a single global invariant function. Pool state is the triple (x, y, w) with w *genuine stored state*: it is in general *not* recoverable as α/x , and the invariant level $k = x^w y^{1-w}$ is a dependent *readout* that trades do *not* conserve.

A trade at strike ray θ with cash leg Δy proceeds as follows.

1. **Trade point.** $T = (x_T, y_T)$ is the unique intersection of the ray $y = \theta x$ with the live curve $x^w y^{1-w} = k$.
2. **Local conserved pair at T .** $\alpha_T = x_T w$, $\beta_T = y_T (1-w)$. The conservation law is applied *at T* , as if T were the reserves point.
3. **Flows, evaluated at T (these are the actual reserve changes):**

$$y' = y + \Delta y, \quad \Delta x = -\frac{\alpha_T \beta_T \Delta y}{(y_T - \beta_T)(y_T + \Delta y - \beta_T)}, \quad \Delta w = \frac{\beta_T \Delta y}{y_T (y_T + \Delta y)}, \quad (3)$$

on the admissible domain $y_T + \Delta y > \beta_T$ (the pole of the closed form) *and* the *global reserve-feasibility* constraints $x + \Delta x > 0$, $y + \Delta y > 0$: the flows are evaluated at T but are drawn from the pool's actual reserves, so feasibility bounds trade size, binding hardest in the deep wings. The cash-leg sign follows the wing/side rule.

4. **Induced global update.** $(x, y, w) \mapsto (x + \Delta x, y + \Delta y, w + \Delta w)$.

Properties, stated plainly.

- **Spot is the special case.** At the reserves point the local pair equals the global pair, so a spot swap conserves global (α, β) and reproduces an ordinary weighted-CFMM swap; this is the only case for which the trajectory hyperbola (2) is the reachable set.
- **Off-ATM trades change global α and β by design.** The per-trade conserved pair is (α_T, β_T) at T , not the global pair; nothing is violated because global (α, β) were never claimed constant across off-ATM trades.
- **w is state, k is a readout.** After an off-ATM trade, $\alpha/x' \neq w'$ in general, so w must be stored; and k is not conserved by trades. (For example, from $(x, y, w) = (10, 10, \frac{1}{2})$ the $\theta = 4$ trade point is $(5, 20)$; $\Delta y = 1$ gives $w' = 11/21$ while $\alpha_0/x' = 22/43$.)

Round-trip residual. The transition rule implies a small *round-trip cost*: opening and then reversing a position at the same ray does not return the pool exactly to its prior state, leaving the trader slightly behind. We position this as path-dependence of the class present in production dynamic-function AMMs (Curve v2’s dynamically re-pegged invariant) and analytically adjacent to loss-versus-rebalancing (Section 9); a quantitative treatment is deferred and flagged as a limitation.

A leg’s notional q is range-invariant: adjusting a position’s strike range reshapes its value and its cash leg but not q . The remaining interface is the notional-to-cash-leg map $q \mapsto \Delta y$ at ray θ : the cash leg is the swap that opens (or closes) the leg’s pool impact, and at ray θ a notional q enters as $\Delta y = q \cdot K(\theta)$ with $K(\theta)$ the dollar strike on that ray (the *at-strike* cash leg—bookkeeping evaluated at the strike ray itself; option pricing enters only at the bought leg, Section 6). This map is the protocol’s at-strike design choice (notional $\times$ strike), not a derived result; the bought-leg sizing interface is carried as partly open in Section 10.

The state-transition rule, with conservation applied at the trade’s own point and w carried as state, rests on a machine-checked spine (LENSKER-NEL `tradeUpdate_alpha/beta/hyperbola` and the constant- m `TemporalAMM invariant/trade_dx`; *trusted-from-prover*, Section 8). The off-ATM trade-point existence/uniqueness lemmas have been submitted to the prover and report complete, but are *not yet folded and manager-audited to our trusted-from-prover bar*; we do not list them among the machine-checked results. The necessity of storing w is shown here by an *arithmetic exhibit* ($w' = 11/21 \neq \alpha_0/x' = 22/43$), not a formalised lemma. We do *not* assert that the pool state is determined by (x, y) alone, that w is a derived field, or that any single function of state is preserved across off-ATM trades; those readings are withdrawn.

3.4 The reference price (“carry”), strike normalisation, and funding

The pool maintains an internal reference price, which we call the *carry* and write $P = N_y/N_x$ —the ratio of the pool’s notional cash holding N_y to its notional underlying holding N_x , i.e. the price the pool treats as its own centre. A strike is registered relative to this carry through the log-distance $u = \log(\text{price}) - \log P$ (how far, in log-price, a given price sits from the carry). Sustained directional drift would otherwise decouple the pool’s spot price from the external oracle spot; *strike normalisation* periodically *rebases* the frame—rescaling the coordinates by a single factor $r > 0$ ($x \rightarrow rx$, $\alpha \rightarrow r\alpha$, $k \rightarrow r^w k$, strike ray $\theta \rightarrow \theta/r$, where θ is the ray identifying a strike, Section 3.3)—bounding drift without nullifying the standing deviation from the anchor curve. Trades and rebases *commute*, so the pricing frame is well defined independent of interleaving (machine-checked: PH6 `R_form_rebase_invariant` and the constant- m `trade_rebase_commute`; *trusted-from-prover*).

Funding anchors pool spot to oracle spot. By analogy with a perpetual future, a position accrues funding at a rate *proportional to the deviation of the live pool curve from the $w = \frac{1}{2}$ anchor curve, evaluated at the position’s own strike ray and read through the lens* (Section 4). The magnitude is the size of the

per-ray, through-the-lens deviation—which carries the lensed exponent $\pm g_{\text{loc}} = \pm m\gamma$, so the kurtosis knob rescales the funding rate by design—and the direction follows the perp analogy (if the pool prices the option above its anchor value, the long side pays). Funding is therefore countercyclical—the crowded side pays the contrarian side—and a rebase bounds the rate at which drift accumulates into the deviation that funding prices.

We are explicit about a property a referee of the prior draft identified: with a *vol-free* mark, this funding *vanishes when the pool sits at its anchor* (the live ray coincides with the anchor ray, so the deviation is zero), so it does not by itself charge a standing put-wing drift at the design’s own equilibrium. The lens is the design’s intended remedy: it makes the mark volatility-calibrated, giving the deviation—and hence funding—a vol-aware scale through $\pm m\gamma$. We do *not* claim here that the lensed funding neutralises any specific equilibrium-drift quantification; that analysis is deferred, and the open question is carried as a limitation.

4 The constant slope-multiplier lens (the kurtosis knob)

The pool above prices coherently within one fixed liquidity shape. The prior draft identified the second free moment—*kurtosis*, the peakedness of the liquidity distribution—as a direction for future work. This draft realises it as a *constant slope-multiplier lens*: a single read-time transformation (we call it a *lens* because every priced quantity is read *through* it, as through a fixed optical element) that rescales the steepness of the option-value curve by one number, without touching the pool’s reserves or trade mechanics.

The lens is the vol-calibration the vol-free mark lacks. A reviewer of the prior (vol-free) draft asked, in effect, for a single static, volatility-calibrated curvature parameter set at pool creation. The constant- m lens *is* that parameter: m is fixed once, from the underlying’s volatility, and never moves under trading. It is the design’s delivered answer to the vol-calibration ask, and it is what gives funding a vol-aware scale (Section 3). We keep the claim modest: the lens supplies a calibrated curvature, not a re-derivation of any option-pricing model, and we do not claim it neutralises any specific equilibrium-drift quantification.

4.1 Definition

Let $\gamma > 1$ be the curve’s convexity exponent (the steepness parameter from the introduction), so that, with S the spot price of the underlying, option value follows the power law value $\propto S^{-\gamma}$ far from the strike (in the *wings*). The lens introduces a single scalar $m > 0$ and sets the *lensed* option-value exponent

$$g_{\text{loc}}(K) = m\gamma \quad \text{for every strike } K, \quad (4)$$

i.e. the steepness is multiplied by m *uniformly across strikes*. Setting $m = 1$ gives $g_{\text{loc}} = \gamma$, the plain Balancer curve; larger m makes the option-value

curve steeper everywhere, and the corresponding trade-point map $\theta_{tx} = \text{mode} \cdot (\text{chosen}/\text{mode})^m$ —which sends a chosen strike ray to where the trade actually lands, measured from the curve’s central (mode) ray—places a trade further out. Here θ_{tx} is the executed ray, chosen the ray the trader selected, and mode the at-the-money centre. Steepness and outward placement therefore move in the *same* direction with m .

What the lens is and is not. The lens is *vol-calibrated at setup and static under trading*: it is the curve’s geometry chosen so the pricing surface suits an asset of a given volatility, not a trader-measured statistic. Crucially the lens leaves the **pool flow unchanged**: the trade rule (3), the oracle-arbitrage map, and the rebase are the plain Balancer dynamics; x , y , and w move exactly as in Section 3. The lens is a forward read applied to everything *read* (pricing, the option chart, settlement, funding, portfolio value) and *written* (trades, settle-at-strike) through one shared helper; it is never inverted, and the underlying pool stays plain Balancer. The pool-byte-identity and the $g_{loc} = m\gamma$ properties are machine-checked (constant- m TemporalAMM: `g_eq_m_gamma`, `g_const_in_strike`, `g_eq_gamma_iff_m_one`, `warp_linear` $\int m = m \Delta\gamma$; *trusted-from-prover*).

The pure-pricing ceiling: level and skew, not smile curvature. We are explicit about how much of an option surface the construction can fit. The cross-strike pricing surface is governed by essentially *two shape parameters*: the tail steepness $m\gamma$ and the skew w , giving a single power-law exponent per wing. It can therefore match a surface’s *level* and *skew*, but not strike-dependent *smile curvature*: a single exponent per wing has no degree of freedom that bends differently across strikes. This is the same object as the single-pool-rigidity limitation (Section 10)—one weight and one lens fix the liquidity shape, which is exactly what buys the closed-form tractability. Whether the single-exponent tail is adequate for a given real asset’s surface is an *empirical (backtest) question*, not one this paper settles; a backtest would speak only to *tail adequacy* and would *not* discharge the carried GH/Bessel-K or Merton hypotheses, which remain conditional regardless. The out-of-the-money liquidation strikes the design targets are level- and skew-dominated, so we expect the smile-curvature ceiling to be second-order there; we state this as a plausibility, empirical and not proven.

4.2 The lens’s natural home (scoped reading, not load-bearing)

There is a clean reading of m as an inverse temperature, which we include only with its scope stated. In the *wings*, where value follows the power law $\text{value}(S) = S^{-m\gamma} = e^{-(m\gamma)q}$ with $q = \log S$ (the log spot price), the value is a Gibbs weight (a Boltzmann-style exponential weight) with inverse temperature $\beta_{th} = m\gamma$ (the “th” marks this as a thermal parameter, distinct from the pool offset β of Section 3), so $\text{value}_m = (\text{value}_1)^m$ and “colder” means “steeper.” This is *wing-scoped*: it describes the power-law tail, not the engine’s *bounded smooth-pasted mark* near the money (where the measured mark-to- S^{-g} ratio departs from unity by orders

of magnitude). m acts as a dilation $\gamma \mapsto m\gamma$, not as a measure tilt. This reading is itself machine-checked as a self-contained statement (LENS_THERMAL: value_is_gibbs, value_pow_m, invtemp_eq_m_gamma, m_one_recovers_base; *trusted-from-prover*) but is *not load-bearing* for the construction: the engine's option value is the bounded smooth-pasted mark, not the bare power law. We note, for completeness and without claiming more, that a *measure-backed* information geometry (exponential family / Fisher metric of a probability measure) is *not* available for the Balancer curve; only a measure-free Hessian / dually-flat geometry survives.

5 American-style exercise: the smooth-pasting free boundary

This is the new core of the paper: the barrier ceiling of the prior draft becomes an early-exercise *free boundary* (a price level not fixed in advance but solved for as part of the problem, marking where the holder optimally exercises) in closed form, with no new parameters.

5.1 The construction

While the holder waits, the position is worth its *continuation value*, which on this curve is $c \cdot s_N$: here s_N is the *normalised strike coordinate* (the strike re-expressed in the curve's own scaled units, so that $\text{value} \propto s_N \propto S^{-\gamma}$) and c is the constant that fixes the overall level of the continuation value. The continuation value *pastes* onto the intrinsic (already-in-the-money) payoff at the free boundary, fixed by *smooth-pasting*—requiring the two pieces to agree in both value and slope, i.e. two conditions in the two unknowns s_N^* and c . The solution is closed-form:

$$s_N^* = \theta \left(\frac{\gamma+1}{\gamma} \right)^\gamma, \quad \text{equivalently } S^* = K \frac{\gamma}{\gamma+1}, \quad (5)$$

$$c = \frac{1}{(\gamma+1) s_N^*}, \quad \text{value fraction at the boundary} = \frac{1}{\gamma+1}. \quad (6)$$

Here $\theta = s_N(K)$ is the strike registered in the curve's carry coordinate; because $\theta \rightarrow \theta/r$ on rebase, s_N^* tracks the strike automatically. The mark reaches its cap of 1 only at *full* exercise, not at the boundary: at S^* the position is worth the fraction $1/(\gamma+1)$.

5.2 The new mark / exercise rule

Per leg, wing-aware, with $s_N = s_N(S)$ and s_N^*, c as above:

- **Continuation** (on the OTM side of s_N^*): $\text{mark} = c \cdot s_N$. This runs *through* the old strike ray; the strike is now an ordinary interior point with no saturation there.

- **Exercise** (past s_N^*): mark = intrinsic from the strike. On the bounded (vertical-spread) wing the intrinsic is $1 - S/K$, reaching 1 at full exercise; on the unbounded (single-option) wing the intrinsic keeps growing past 1 and is *not* capped (an uncapped option is correct; bounding comes from using a spread).
- **Trigger**: the ratio intrinsic/continuation reaches 1 exactly at s_N^* , so that crossing is the exercise switch and it lands at the formula point, not at the money.

This generalises the barrier rule of the prior draft. There, the mark saturated to 1 at the strike and stayed at the ceiling once ITM; here the ceiling is lifted into the free boundary (5), and the closed-form continuation runs past the strike before exercise pays intrinsic. The curve functions, calibration, and funding are unchanged by the rule—it lives entirely in the per-leg mark.

5.3 A worked example (illustrative numbers)

To show the formulas in motion we evaluate them at *illustrative* parameter values—chosen for arithmetic clarity, *not* calibrated to any market or asset, and not an empirical validation of anything. Take a put leg (hedging a long) with convexity exponent $\gamma = 2$, dollar strike $K = \$100$, and lens $m = 1$ (plain Balancer, so $g_{\text{loc}} = m\gamma = 2$). The early-exercise free boundary (5) sits at

$$S^* = K \frac{\gamma}{\gamma + 1} = 100 \cdot \frac{2}{3} = \$66.67,$$

below the strike (the put becomes optimal to exercise only after spot has fallen well past K), and at that boundary the mark equals the value fraction (6), $1/(\gamma + 1) = 1/3$. On the OTM side ($S > S^*$) the continuation mark is cs_N with $s_N \propto S^{-\gamma}$, so it scales as $\frac{1}{\gamma+1} (S^*/S)^\gamma$, growing as spot falls toward the boundary. Steepening the lens to $m = 3$ (so $g_{\text{loc}} = 6$) leaves S^* unchanged (the boundary (5) depends on γ , not m) but makes the off-boundary mark decay far faster:

spot S	mark, $m = 1$ ($g_{\text{loc}} = 2$)	mark, $m = 3$ ($g_{\text{loc}} = 6$)
$\$66.67 = S^*$	0.333	0.143
$\$80$	0.231	0.048
$\$90$	0.183	0.024
$\$100 = K$	0.148	0.013
$\$120$	0.103	0.004

The table simply traces (5)–(6) and the wing power-law value $\propto S^{-m\gamma}$ at these chosen inputs; it illustrates the closed forms (and the kurtosis knob’s effect on tail steepness) and makes no empirical or market-calibration claim.

5.4 Why this is the American boundary

The smooth-paste solution (5)–(6) is not an arbitrary C^1 join. It is the *unique* solution of the value-match plus slope-match system, and it coincides with the optimal early-exercise boundary in the variational sense. Both facts are machine-checked: the boundary is *generated* as the unique smooth-pasting solution (T1a `Sstar_A_forced`, `coeffA_forced`; *trusted-from-prover*), and S^* maximises the holder’s value over *deterministic* exercise boundaries (T1b `opt_boundary_is_max_A`, a genuine variational result, *trusted-from-prover*). The identification with the stochastic optimal-stopping (Snell) boundary is *named but not formalised*—the `AmericanOptimalityPrinciple` field is a placeholder obligation (`True`), not a discharged premise. The closed-form boundary also sits inside a deeper tie to the classical perpetual-American (Merton) problem. Here r is the risk-free interest rate and σ the underlying’s return volatility (this r is distinct from the rebase factor of the same name). The exponent γ is the characteristic root of the underlying’s log-price Laplace exponent. Under the *symmetric pairing* we use—a put root $-\gamma$ and a call root $\gamma + 1$, equivalently zero net carry between the two legs, so that the two log-price roots sum to 1—the product-of-roots (Vieta) relation reads $\gamma(\gamma + 1) = 2r/\sigma^2$, which holds in the regime $r > \sigma^2$; this is the convention our Lean `merton_vieta_sum/merton_vieta_prod` and the FAITH-4 `faith_merton` gate both encode. (The classical no-dividend Black–Scholes–Merton slice, by contrast, keeps only the single decaying root and gives the *linear* relation $\gamma = 2r/\sigma^2$; we do not use that slice here.) On the symmetric pairing, S^* is the Merton boundary (MERTON tie `Sstar_is_merton_boundary`, `gaussian_limit_quadratic`; *trusted-from-prover*, with the GH/Bessel-K distributional layer carried as named hypotheses).

5.5 The C^1 seam, machine-checked two ways

At s_N^* the continuation and intrinsic must agree in value *and* slope. This is checked two ways. *Formally*: the value-match and slope-match conditions are machine-checked at the boundary for every steepness $g > 0$ (R1 `valueMatch_A/slopeMatch_A`; `LENSKERNEL valueMatch_g/slopeMatch_g` for arbitrary g , hence directly at $g = m\gamma$; `constant-m paste_value/paste_slope`; *trusted-from-prover*), and the OTM→ITM crossover lands at the dollar strike K for all γ under the registration $\theta = s_N(K)$ (R2 `crossover_sNorm_at_K`; *trusted-from-prover*). *On the engine*: the live engine satisfies the seam to 0.04% in value and 0.1% in slope for $\gamma \in \{1.5, 2, 3, 4\}$ (the 0.1% is the curve table’s $S^{-\gamma}$ accuracy, not a structural gap), under an acceptance gate of $\leq 0.15\%$ on both with no jump and no kink. A companion no-jump check confirms the held position’s value is continuous across the at-the-money / OTM↔ITM crossing on the live value path (a distinct locus from the settlement seam; this is an engine gate, not a Lean theorem).

5.6 The replication scope: $|I| \leq 1$ exact, $|I| > 1$ approximate

The construction replicates the target American payoff *exactly* for delta magnitude $|I| \leq 1$. This bound is *per wing*: each wing manufactures a delta of magnitude at most one from reserves, so on its own wing the replication is exact. The shipped instrument is a vanilla *opposite-wing band*, and its defining clause—two legs on opposite wings, with at most one in-the-money at any time (Section 6)—*is* the per-wing guarantee: because spot cannot be in-the-money on both wings at once, the active wing is always a single $|I| \leq 1$ leg. Hence for the shipped product replication is exact.

The mutually-exclusive regime, $|I| > 1$, arises only outside that design: under same-wing stacking (multiple legs accumulating delta on one wing) or single-leg leverage, where “true American” exercise and “exact replication per wing” cannot both hold. This is excluded by the opposite-wing band design and we therefore scope it out of the shipped product; but we keep it *disclosed* as a genuine limitation: outside the opposite-wing band, $|I| > 1$ is a *labelled approximation*, and we do not claim exact replication there (Section 10).

6 Settlement

A perpetual-option position is opened as a *band*—a composition of option legs—against a specific perpetual future the trader already holds, the *origin perp*. When the band is opened it is minted against a carved slice of that aggregated perp; the slice’s absolute notional, equity, and entry mark are frozen for the life of the band, so that settlement attribution cannot be contaminated by unrelated perp activity. Settlement is therefore *perp-denominated*: leg values are carried in units of the carved perp and converted to cash at the slice’s closing equity, with the band’s frozen leverage L_0 amplifying the settled net and a counterparty-equity floor as a final guard.

Closing and exercising are economically equivalent for the holder. Each piece of a position carries its *original strike* (as opened) and an *effective strike* that equals the original while OTM. Under the American rule the effective strike is simply the original strike throughout—continuation and exercise are both referenced to K —so the closing swap returns the position’s full value, intrinsic included when exercised, with no separate exercise path. The composite-ray closed form for settlement holds across the OTM-to-ITM boundary under this substitution (machine-checked: C1 `compositeRay_ITM_substitution`; *trusted-from-prover*). A band has two legs on opposite wings, and since spot cannot be ITM on both at once, at most one leg is ITM at any time; settlement then has a clean two-case structure (both legs reversed on the AMM when neither is ITM; the ITM leg settled to cash by the closed form and the OTM leg reversed when one is). The stage-2 mark/exercise change does not alter the stage-3 carved-perp→dollar conversion, which applies uniformly to continuation and exercised legs.

7 No internal arbitrage

The sharp statement we can make concerns the *costless collar*: a band whose sold and bought legs are arranged to be cash-neutral by construction. We deliberately do *not* claim that “no composition of swaps across rays yields a riskless edge”—no formal result covers swap compositions, and that prose claim is withdrawn until one does. What we establish is a symmetry of a posited collar surplus, at the strength the derivation actually supports.

Proposition 1 (Collar surplus vanishes exactly at symmetry). *The costless-collar capital-efficiency surplus is zero if and only if the pool is symmetric, i.e. $w = \frac{1}{2}$. Under skew ($w \neq \frac{1}{2}$) the surplus is non-zero, with an explicit counterexample.*

The biconditional is the substance, and we state its strength honestly. The Lean `collarSurplus` is a *posited* structural form, $\text{collarSurplus}(\theta, w) = \theta((1 - w)/w - 1)$, proportional to anchor asymmetry; its zero-set is θ -independent, so the $\forall\theta$ quantifier does no work, and the result is a *symmetry statement about a posited surplus*, not a mechanism-level no-arbitrage theorem quantifying over swap compositions. It is not (yet) derived from the $\min(\text{slope}, 1/\text{slope})$ mark formula; we carry the surplus form as a named definition and exhibit the skew sign explicitly. The discrimination between symmetric and skewed pools is genuine, and the edge is a *skew phenomenon*—it exists only when the pool is directionally skewed and vanishes exactly at symmetry. The statement is a claim about *pricing*, not about the pool being left unchanged: a costless collar’s two legs transact in the same direction against the curve, so the pool does move (w shifts), but the trader pays for that displacement at fair, sequenced prices (the proceeds of a swap are the path integral of the marginal price along the curve), so there is no free cash to extract. Liquidation protection retains genuine risk-management value even at zero skew; zero skew removes the *free* capital-efficiency edge, not the *worth* of the protection.

Proposition 1 is machine-checked (C2 `collarSurplus_zero_iff_half`; *trusted-from-prover*, with the surplus form carried as a named definition). A companion result formalises the no-arbitrage symmetry as a reflection: the put at strike coordinate s is the call reflected through the at-the-money ray (the strike $s \mapsto \theta^2/s$, θ the centre ray), so no-arbitrage is the curve’s symmetry (C3 `reflection_arrow`; *trusted-from-prover*). We are precise about its status: the reflection *arrow* is a proved identity, but the link from the spec mark to the live engine’s instrument remains a residual modeling premise (the engine-faithfulness question), so C3 is best read as a *conditional skeleton* rather than an unconditional no-arbitrage theorem over the production engine.

8 Formal verification and the open frontier

8.1 Provenance discipline

All Lean results below were compiled by the Aristotle theorem prover server-side in the matching toolchain (Lean 4.28.0 / Mathlib v4.28.0) and

passed a zero-cost artifact audit: a forbidden-token scan (no `sorry/admit/native_decide/sorryAx/opaque/unsafe`), the prover-reported `#print axioms` restricted to `{propext, Classical.choice, Quot.sound}`, an unscoped-module byte-diff, and a hand re-derivation of the statements. They were *not* re-run against a local canonical kernel in this work; we therefore label them *trusted-from-prover* and never “verified.” A local canonical-env build is the single label upgrade to “verified” and is environment-blocked here. We also distinguish *grounded* results (proved over the real objects with no load-bearing carried hypothesis) from *carried* results (conditional on named structure-field hypotheses that are not discharged, never on `axiom` declarations).

Inspecting the artifacts. The Aristotle prover is a private server and is not itself publicly reproducible. The Lean *statements and proof source*, however, are available in the project repository under `formal/` (the Lean project and its provenance map `formal/INDEX.md`; per-result archives under `aristotle_runs/`), so a reader can read and check the exact theorem statements and proof scripts even though the prover is not public. We stress that this does *not* upgrade the label: the artifacts were compiled by Aristotle and remain *trusted-from-prover* (Aristotle-compiled, audited, but not re-run against a local canonical kernel in this work). A reader can inspect what is claimed; a local canonical-env rebuild is the separate step that would license “verified,” and it is environment-blocked here.

8.2 What is machine-checked (*trusted-from-prover*)

- **Settlement free boundary.** C^1 smooth-pasting at $S^* = K\gamma/(\gamma + 1)$ on both wings (R1 `valueMatch_A/slopeMatch_A`, grounded), generated as the *unique* solution (T1a `Sstar_A_forced`, `coeffA_forced`, grounded), and as the *optimal* exercise boundary over *deterministic* boundaries (T1b `opt_boundary_is_max_A`, variational; the stochastic Snell identification is named but not formalised). The crossover lands at the dollar strike for all γ (R2 `crossover_sNorm_at_K`, grounded). The seam holds for arbitrary steepness $g > 0$, hence at $g = m\gamma$ (LENSKERNEL `valueMatch_g/slopeMatch_g`; `constant-m paste_value/paste_slope`, grounded).
- **Merton tie.** γ is the characteristic root of the log-price Laplace exponent. Under the symmetric put/call-root pairing (put root $-\gamma$, call root $\gamma+1$, root-sum 1), the Vieta product relation reads $\gamma(\gamma + 1) = 2r/\sigma^2$ (regime $r > \sigma^2$), with r the risk-free rate and σ the asset volatility; S^* is the Merton boundary (MERTON tie, grounded in the algebra/curvature/limit/boundary; carried on the GH/Bessel-K distributional layer). The classical no-dividend Black–Scholes–Merton slice instead gives the single decaying root $\gamma = 2r/\sigma^2$ and is not the convention used here. The GH asymmetry is recorded honestly: only the put eigenfunction is native (the call root leaves the strip).
- **Curve / pool spine and lens.** The pool flow, w -skew rule, conservation hyperbola, and rebase commutation (`constant-m TemporalAMM`:

invariant, trade_dx, trade_rebase_commute, price_is_grad, R_psd; PH6 R_form_rebase_invariant; LENSKERNEL pool rows). The lens facts $g_{loc} = m\gamma$ constant in strike, $m = 1 \Leftrightarrow$ plain Balancer, the round-trip trade map, and the linear warp $\int m = m \Delta\gamma$ (constant- m g_eq_m_gamma, g_const_in_strike, g_eq_gamma_iff_m_one, thetaTx_roundtrip, warp_linear, all grounded).

- **No internal arbitrage.** The collar biconditional (Proposition 1, C2; carried on the surplus form) and the reflection arrow (C3; arrow grounded, engine link residual).
- **Lens thermal reading** (Section 4.2), wing-scoped and non-load-bearing (LENS_THERMAL, grounded).

8.3 The conditional-solvency interface stack

A separate prover line formalises the curve $\rightarrow$ value $\rightarrow$ storage interface and its passivity/solvency guarantees as a typed stack (AMMCurve.lean, Temporal.lean, Seam.lean). It establishes, machine-checked: (i) a *curve gate*—a shape failing convexity/monotonicity/coercivity cannot be constructed, so cannot reach any theorem; (ii) *short-gamma universality*—pool value is concave in price for any valid curve (poolValue_concave0n) and this propagates into stored equity (intrinsic_concave0n) by types, not inspection; (iii) the headline **reserves_have_no_floor**: with a convex obligation $O = p^2$, reserve value is unbounded below, so *the solvency floor cannot come from reserves*—“convexity must be funded,” as a theorem at the storage interface; and (iv) conditional solvency: *given* a port that covers the concave deficit, a curve-wired pool is passive and solvent for all time with no free lunch (CurvePool.solvent_forever). All *trusted-from-prover*; the proven instances are cpmm (equal-weight Balancer frontier) and expPool (exponential).

The open frontier, stated plainly. Solvency is *conditional*. The hypotheses B1 (the port covers the floor), B3 (arb is one-way), and B4 (the ledger closes) are *structure fields*, not discharged facts. The stack proves “*if* the ledger closes, arb is one-way, and the port covers the floor, *then* passivity and solvency”; it does *not* prove the port pays in the real world, which is the external funding/hedge assumption. **reserves_have_no_floor** shows the port is *necessary*, not *sufficient*: geometry is necessary-never-sufficient for solvency. We do not claim solvency is closed by the construction.

What the stack does not establish. The typed contracts (price/tick/ledger/portFlow/arbLeak) are *not* wired to the production HTML engine’s numeric formulas; the package machine-checks the interface and its propagation, not that the engine instantiates the fields. That instantiation (engine formulas $\vdash$ B1/B3/B4) is the next body of work. The instances are representative, not exhaustive: weighted Balancer and the power-sum family are not instantiated, only the mechanism that would transfer through the abstract interface.

8.4 The whole-exchange “single object” (research direction)

We have a partial result toward unifying the entire exchange—storage, flow, gauge, funding port, settlement port, dollar pipe, and a parameterised close mechanic—as a *single* port-Hamiltonian object whose solvency reads as passivity-under-admissible-inputs. The *internal* half is machine-checked (*trusted-from-prover*): `exchange_internal_passivity`, with the resistive curvature’s nonnegativity discharged *by the geometry* (the proven `R_psd`), so the system is passive *because* the geometry is PSD, plus `trade_no_spontaneous_storage` and `no_internal_free_money`. The *external*/solvency half stays *open*: `solvency_of_coverage` keeps the coverage premise (B1/B3/B4 admissibility) as an undischarged hypothesis, honoring `reserves_have_no_floor`. We therefore present the unified single object as a *conjecture / research direction*—internal-passivity half *trusted-from-prover*, external/solvency half *open*—and *not* as a fully unified proven object. (This object’s potential is the Balancer constant-product potential, not a GH cumulant-generating function.)

9 Related work

The construction sits between constant-function market makers and perpetual options. The weighted constant-product invariant is from Balancer [6], whose Liquidity Bootstrapping Pools already use *time-varying weights*; our per-trade weight update is a finer-grained, trade-driven analogue of that time-varying-weight history, not a new curve family. The broader CFMM theory and pricing-oracle properties are from [4,5], with Uniswap v2/v3 as the reference concentrated/constant-product designs [2,3].

Translated CFMMs and replication (the deflation). The observation that each of our trades is, under shifted virtual reserves, an ordinary constant product is exactly the translated/displaced-curve perspective of Evans [8]; the replicating-payoff theory of Angeris, Evans, and Chitra [9], instantiated concretely in RMM-01 [10], shows how to choose a CFMM trading function to replicate a target payoff. We therefore do *not* claim the curve family as a contribution: the geometry is a member of that known design space. The contribution is the per-trade endogenous weight update serving a tradable per-strike claim and the semantic layer (Section 3).

Concentrated-liquidity perpetual options and dynamic weights. InfinityPools [11] builds leverage/options-like exposure on concentrated liquidity ranges, and Panoptic [7] is the closest oracle-free perpetual-options protocol; our design differs in pricing the whole OTM continuum from one pool state and in lifting the barrier into an American free boundary. QuantAMM’s “Temporal Function Market Making” is *prior* work on weight-dynamic AMMs (not concurrent); we treat it as prior art and accordingly do not adopt the “TFMM” name—we use a neutral “per-trade weight-dynamic AMM” descriptor for the mechanism, to avoid the naming collision.

Formal-methods AMM lineage. The Lean-AMM line is anchored by Pusceddu and Bartoletti [12] (FMBC 2024), which formalises AMM properties in a proof assistant; our formal-verification ledger (Section 8) extends that lineage to the smooth-pasting boundary, the lens, and the conditional-solvency interface stack. The formal-methods backing rests on Lean 4 [14] and Mathlib [15]; the perpetual-American (Merton) tie connects to the classical optimal-stopping treatment of perpetual American options.

Loss-versus-rebalancing and the round-trip cost. The nearest analytic neighbour reads an AMM LP position as optionality: loss-versus-rebalancing (LVR) of Milionis, Moallemi, Roughgarden, and Zhang [13] quantifies the LP’s adverse-selection/convergence cost, and the LP’s instantaneous exposure is naturally read as a continuum of (perpetual) short-option positions—the *dual* of what we do, since we *price and sell* a perpetual-option continuum from the same curve geometry whose LP exposure that line analyses. We also use the LVR framing for the *round-trip residual* of our state-transition rule (Section 3): opening and reversing at the same ray leaves a small trader-borne round-trip cost, a path-dependence of the same class present in production dynamic-function AMMs (e.g. Curve v2’s dynamically re-pegged invariant). A quantitative treatment of that residual against the LVR convergence cost is deferred.

10 Limitations

We state the boundaries of the present work plainly, sorted into two kinds: *scope* (boundaries of what we deliberately built—not defects) and *conditional / open* (the price of the headline, where a claim rests on an undischarged hypothesis or an unfinished surface). Several are restatements of the honest-claims posture above; none is dropped in the regrouping.

10.1 Scope (boundaries of the design, not weaknesses)

- **Replication scope is $|\Gamma| \leq 1$, per wing.** Exact replication holds for delta magnitude $|\Gamma| \leq 1$, which is a per-wing bound that the opposite-wing band design guarantees on the active leg (Section 5.6). The $|\Gamma| > 1$ regime—a labelled approximation—arises only under same-wing stacking or single-leg leverage, both excluded by the shipped opposite-wing band; we scope it out of the product but keep it disclosed as a genuine limitation outside that design.
- **Single-pool rigidity (and the pricing ceiling).** The design exposes one directional weight and one rebasing frame, with no per-strike LP discretion; this rigidity is the source of the closed-form tractability and the structural no-arbitrage property, but fixes the liquidity shape up to the lens m . Concretely the cross-strike surface is $\sim$ two shape knobs (tail steepness $m\gamma$, skew w ; one exponent per wing), so it fits surface level and skew but not strike-dependent smile curvature (Section 4); tail adequacy for a real asset is an

empirical (backtest) question. Calibration of funding elasticity and related parameters is protocol-specific and left to implementation.

- **No measure-backed information geometry.** The kurtosis-knob’s thermal reading is wing-scoped and non-load-bearing; only a measure-free Hessian geometry is available for the Balancer curve. This is a boundary on an optional interpretive reading, not on the construction.

10.2 Conditional / open (the price of the headline)

- **Solvency is conditional—the headline open item.** The solvency guarantee holds only under the funding-port hypothesis (B1), with B3/B4 as further named hypotheses. Geometry is necessary (`reserves_have_no_floor`) but never sufficient. The port is not discharged here; that it pays in practice is an external assumption. This is a binary open question of a different kind from the proof-status and basis-point-scale items below: it is the principal direction for future work, not a refinement.
- **Engine $\neq$ spec: the bridge is an oracle, not an extraction.** The trusted-from-prover interface stack is not wired to the production engine’s numeric formulas; the bridge from the live HTML kernel to the Lean definitions is currently an oracle (testing) bridge, not a verified extraction. No claim that “the HTML is Lean-verified” is made.
- **Arbitrary-asset extension rests on a GH/Bessel-K layer.** The Merton tie is exact on the Gaussian slice; extending the boundary to a non-Gaussian asset carries a GH/Bessel-K distributional layer as *named hypotheses* (Section 5), with a recorded asymmetry (only the put eigenfunction is native; the call root leaves the strip). These hypotheses are carried, not discharged.
- **Provenance is *trusted-from-prover*, not “verified.”** Every Lean result is Aristotle-compiled and audited but not re-run against a local canonical kernel; the upgrade to “verified” awaits a local build.
- **The unified single object is a direction.** The internal-passivity half is *trusted-from-prover*; the external/solvency half is open. The whole-exchange port-Hamiltonian object is a conjecture, not a finished proven artifact.
- **Settlement-ledger conservation is open.** The carved-perp settlement (Section 6) is an open design surface: who funds the leverage wedge, the negative-net branch, a magnitude-aware counterparty floor, and the carved-equity boundary at liquidation are not all closed here. We carry these as open rather than imply a balanced ledger.
- **Momentum-band orientation / unpriced convexity is open.** Whether a bought wing must oppose the origin perp (so that band convexity is paid for rather than extracted) is an unspecified product-level decision; until it is restricted or priced, it is a limitation.
- **Settle-sandwich cost floor needs a fee model.** With no fee model specified, no manipulation-cost floor on settlement timing is claimed.
- **Round-trip residual.** The state-transition rule leaves a small trader-borne round-trip cost (Section 3); we position it as path-dependence à la Curve v2 and LVR [13], with a quantitative treatment deferred.

- **Citation provenance.** The prior-art references added in this revision (the arXiv identifiers for Evans 2020, Angeris–Evans–Chitra 2021, Pusceddu–Bartoletti 2024, Milionis et al. (LVR), and the Panoptic authorship correction) are sourced from the prior referee report and have *not* been independently re-verified against arXiv in this draft; every reference will be hand-checked against primary sources before submission.
- **Notional-to-cash-leg interface.** The $q \mapsto \Delta y$ map is stated for the at-strike cash leg (Section 3.3); a full treatment of position-opening pool impact across the bought leg’s pricing is folded into the settlement design and remains partly open.

11 Conclusion

A single liquidity pool on the weighted constant-product curve with one dynamic weight—run as a state transition system whose conservation law is applied at each trade’s own point—prices the entire OTM perpetual-option continuum in closed form. This draft lifts the prior barrier instrument into an American-style perpetual option: a closed-form smooth-pasting free boundary $S^* = K\gamma/(\gamma + 1)$ with a C^1 seam and no new parameters, backed by machine-checked (*trusted-from-prover*) settlement, boundary-uniqueness, optimality, and Merton-tie results, and a constant slope-multiplier lens m that acts as a vol-calibrated kurtosis knob while leaving the pool flow unchanged. On internal arbitrage we claim only what is proved: a posited costless-collar surplus vanishes iff the pool is symmetric—a symmetry statement, not a no-arbitrage theorem over swap compositions, which we withdraw pending a result that covers them. The honest frontier is solvency: geometry is necessary (**reserves_have_no_floor**) but never sufficient, and the solvency guarantee remains *conditional* on a funding port that is shown necessary but not discharged. Closing that port hypothesis—and wiring the trusted-from-prover interface to the production engine—are the principal directions for future work.

A Notation

Symbol	Meaning
x, y	pool reserves (amount of the underlying asset, amount of cash)
S	spot price of the underlying
w	dynamic weight ($w \in (0, 1)$); sets how the curve leans between the assets
k	pool invariant $k = x^w y^{1-w}$ (not a depth measure across rebases)
α, β	curve offsets (“conserved quantities”) $\alpha = xw$, $\beta = y(1 - w)$
u, v	shifted virtual reserves $u = x - \alpha$, $v = y - \beta$ (note: u also denotes the carry log-distance $\log(\text{price}) - \log P$, by local definition)
$T; x_T, y_T, \alpha_T, \beta_T$	trade point on a ray, and the local quantities evaluated there
θ	strike ray (slope of the ray $y = \theta x$), $\theta = s_N(K)$; $\theta = 1$ is at-the-money
θ_{tx}	executed ray = mode $\cdot$ (chosen/mode) m (chosen = selected ray, mode = at-the-money centre)
K	dollar strike price
q	position notional, range-invariant (separately, $q = \log S$ locally in Section 4.2 only)
$\Delta x, \Delta y, \Delta w$	per-trade changes in reserves/weight (Δy = cash leg)
mark	position value as a fraction of one perpetual future, $\min(\text{slope}, 1/\text{slope}) \in (0, 1]$
Γ	hedge ratio (position sensitivity to spot); exact replication needs $ \Gamma \leq 1$ per wing
γ	curve convexity (steepness) exponent, $\gamma \in (1, 4)$
m	constant slope-multiplier (kurtosis/vol knob), $m > 0$
g_{loc}	lensed option-value exponent = $m\gamma$
s_N	normalised strike coordinate, $s_N = (1 - w)/w$ at the 45° point (pinned convention)
s_N^*, S^*, c	smooth-pasting free boundary ($S^* = K\gamma/(\gamma + 1)$) and continuation-value coefficient c
r	rebase factor ($r > 0$). <i>Distinct</i> from the risk-free interest rate, also written r , that appears only in the Merton relation $\gamma(\gamma + 1) = 2r/\sigma^2$ under the symmetric root-pairing, regime $r > \sigma^2$ (Section 5)
σ	asset return volatility (Merton symmetric-pairing relation only)
β_{th}	thermal inverse temperature = $m\gamma$ (Section 4.2; distinct from the offset β)
L_0	position leverage, frozen at band-open
P	carry (the pool’s internal reference price) $P = N_y/N_x$
N_x, N_y	pool notional holdings of underlying and cash (define the carry P)

B Formal-verification artifact map

Each row cites the headline theorem(s), the result’s provenance label, and the returned archive. All rows are *trusted-from-prover* (Aristotle-compiled, audited, not locally re-run). “grounded” = no load-bearing carried hypothesis; “carried[h]” = conditional on the named undischarged structure-field hypothesis h .

Headline theorem	Depth	Archive (returned .lean)
valueMatch_A/slopeMatch_A (R1)	grounded	aristotle_runs/R1
crossover_sNorm_at_K (R2)	grounded	aristotle_runs/R2
compositeRay_ITM_substitution (C1)	grounded	aristotle_runs/C1
collarSurplus_zero_iff_half (C2)	carried[surplus form]	aristotle_runs/C2
reflection_arrow (C3)	grounded (arrow); residual engine link	aristotle_runs/C3_reflection
Sstar_A_forced/coeffA_forced (T1a)	grounded	AIRTIGHT_T1a_invert
opt_boundary_is_max_A (T1b)	grounded (variational); carried[Snell]	AIRTIGHT_T1b_optimality_clean
single_source/price_is_grad/Rgsnd (T2)	grounded	AIRTIGHT_T2_singlecore
Sstar_is_merton_boundary (MERTON)	grounded; carried[GH/Bessel-K]	MERTON_tie
valueMatch_g/slopeMatch_g (LENSKERNEL)	grounded ($\forall g > 0$)	aristotle_runs/LENSKERNEL
g_eq_m_gamma/paste_value/warp_gamma (constant- m)	grounded	aristotle_runs/MONOLITH_CONSTM
value_is_gibbs/invtemp_eq_m_gamma (LENS_THERMAL)	grounded (wing-scoped)	aristotle_runs/LENS_THERMAL
poolValue_concaveOn/reserves_have_dofloor	grounded	Seam.lean
CurvePool.solvent_forever	carried[B1/B3/B4]	Seam.lean
exchange_internal_passivity	grounded (internal half)	PH_UNIFICATION_COMPOSED
solvency_of_coverage	carried[coverage] (OPEN external)	PH_UNIFICATION_COMPOSED

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